

When Should a Rule Learn? Predicting the Rule / Neural-Net / ML Composition for Classification Streams

Anonymous authors

Paper under double-blind review

Abstract

A production classification site can be served by a hand-written rule, a (neural-net) LLM call, or a trained classical model. Common practice treats the rule→LLM→ML progression as a fixed ladder whose destination is always the trained model. We ask a data-centric question instead: **given a stream’s measurable characteristics, which of these tiers should serve it, and when?** On 19 public text datasets (plus preliminary image and audio), scored apples-to-apples on a common test set across four tiers (keyword rule, zero-shot LLM, few-shot LLM, classical-ML training curve), we find a clear, interpretable gradient: **label cardinality and rule-accuracy-over-chance predict which tier wins** — high-cardinality streams favor the trained classical model, low-cardinality/binary streams favor the LLM. We operationalize this as **Adaptive Graduated Autonomy (AGA)**, which selects the *terminal* tier per stream (often the LLM, not the trained model) under a *multi-objective* policy — modality feasibility and a latency SLO as hard constraints, accuracy within a significance band, operating cost minimized — and graduates under a statistical non-inferiority gate. This paper measures the accuracy, cost, AND latency axes (modality feasibility is first-class but a hard filter, not a measured scalar). AGA matches the fixed-ladder baseline’s accuracy while consuming **~44% fewer labeled outcomes**, and keeps the LLM as terminal on roughly half of streams. We report two honest negatives that strengthen rather than weaken the picture: a *learned cross-site predictor* of the best tier does **not** generalize at this fleet size (leave-one-dataset-out below the majority baseline) — per-site measurement is required — and a vision-LLM-on-spectrogram proxy fails on environmental audio. The contribution is the characterization and the cost-reducing adaptive algorithm, not the cascade or the gate (both prior art).

1 Introduction

(Keep the “every production codebase has classify functions” motivation; drop product language.)

What is *not* claimed as novel: LLM cost-cascades (FrugalGPT) and learning-to-defer route between models already; paired-McNemar classifier comparison is standard. What *is* contributed: (1) a characterization of which dataset characteristics predict the cost-optimal tier, (2) an adaptive algorithm that exploits it to cut labeled-data cost at parity accuracy, and (3) honest evidence on where a learned cross-site policy does and does not work.

1.1 Contributions

1. **Aligned multi-tier benchmark** — rule / zero-shot LLM / few-shot LLM / classical-ML-curve, all on one common test set per dataset, on a shared labeled-outcomes-consumed axis, across 19 text datasets (sentiment, intent, moderation, emotion, topic, spam) + preliminary image/audio.
2. **Predictive characterization** — label cardinality (Spearman $\rho \approx -0.7$ vs outcomes-to-graduate) and rule-over-chance predict which tier wins and how deep ML must train.
3. **AGA** — chooses the terminal tier and graduates under a non-inferiority gate; matches fixed-ladder accuracy at ~44% fewer labeled outcomes.

4. **Two honest negatives** — a learned cross-site tier-predictor does not generalize at 21 datasets; spectrogram-vision fails on audio. Plus a tunable non-inferiority (“close-enough”) gate for cost-aware graduation.

2 Setup

- **Tiers.** RULE $\rightarrow$ MODEL (LLM, zero/few-shot) $\rightarrow$ ML (classical sklearn head). The LLM is the neural-net tier; ML is the trained classical tier.
- **Datasets.** 19 text + CIFAR-10 (image) + ESC-50 (audio).
- **Aligned protocol.** Common test subset per dataset (text $n=400$); rule from a 100-example seed; ML as a budget curve; LLM (Claude Haiku) zero- and few-shot (≤ 40 exemplars). Each tier placed at its true labeled-outcome cost (rule ≈ 100 , zero-shot=0, few-shot=k, ML=budget).
- **Two costs.** We separate **one-time labeled-outcome cost** (to train ML / seed the rule) from **perpetual per-call inference cost** (the LLM bills every call forever; rule/trained-ML are ~free per call). AGA’s savings are reported on the labeled-outcome axis; the steady-state inference-cost argument is what makes graduating *off* the LLM valuable.

3 The composition varies by stream

Sorted by label cardinality, the gradient is visible: high-cardinality $\rightarrow$ trained ML; binary/ low-cardinality $\rightarrow$ LLM. (esc50 ML “wins” only because the spectrogram-vision proxy fails; cifar10’s LLM “win” reflects a weak pixel-ML baseline — see §6.)

dataset	labels	rule	ML	MODEL-zs	MODEL-fs	best
clinc150	151	0.172	0.838	0.748	0.725	ML
banking77	77	0.225	0.870	0.777	0.807	ML
hwu64	64	0.247	0.805	0.850	0.865	MODEL-fs
esc50	50	0.058	0.283	0.017	0.050	ML
atis	26	0.733	0.848	0.598	0.815	ML
twenty_newsgroups	20	0.133	0.660	0.685	0.680	MODEL-zs
dbpedia14	14	0.632	0.968	0.968	0.978	MODEL-fs
codelang	12	0.892	0.986	1.000	1.000	MODEL-zs
cifar10	10	0.173	0.327	0.780	0.340	MODEL-zs
yahoo_answers	10	0.240	0.670	0.740	0.720	MODEL-zs
snips	7	0.755	0.983	0.970	0.983	ML
emotion	6	0.323	0.818	0.632	0.608	ML
trec6	6	0.420	0.855	0.795	0.887	MODEL-fs
ag_news	4	0.375	0.895	0.853	0.892	ML
tweet_emotion	4	0.403	0.613	0.828	0.810	MODEL-zs
imdb	2	0.532	0.853	0.938	0.958	MODEL-fs
rotten_tomatoes	2	0.487	0.723	0.930	0.940	MODEL-fs
sms_spam	2	0.920	0.945	0.887	0.988	MODEL-fs
sst2	2	0.512	0.770	0.968	0.978	MODEL-fs
tweet_hate	2	0.497	0.560	0.770	0.640	MODEL-zs
tweet_offensive	2	0.660	0.775	0.752	0.730	ML

4 Adaptive Graduated Autonomy

- **oracle_terminal** is *multi-objective*, applied in priority order: (i) **modality feasibility** (hard filter — e.g. spectrogram-vision can’t read words), (ii) **latency SLO** (hard — a tier breaching the real-time budget is out), (iii) **accuracy** within a Wilson-CI significance band, (iv) **operating cost** minimized within the band (perpetual inference \$ + labeled cost). The LLM wins only when *significantly* better on accuracy *and* feasible within the latency/modality constraints; otherwise a rule/trained-ML tie

graduates off the LLM. Cost is the axis this study measures; latency and modality are first-class and default to unconstrained.

- **Tie-to-graduate:** ML graduates once it *ties* the LLM (non-inferiority), not once it beats it.
- **CostToleranceGate:** the non-inferiority gate; ϵ is operator-tunable.
- **The router is permanent:** it holds the rule floor, watches for drift, and re-escalates to the LLM when the data demands — only the LLM *dependency* shrinks.

4.1 Measured latency + cost (the steady-state axes)

Per-call latency (p50): rule 0.003 ms, classical-ML 0.1 ms, MiniLM-embed ML 6-23 ms, LLM 530-570 ms — the rule is ~200,000x faster than the LLM, classical ML ~5,000x. LLM per-call cost \$0.0001-0.0006 (scales with label-prompt length); rule/ML ~\$0. Under a real-time SLO (e.g. 50 ms) the LLM is structurally excluded, so latency *forces* graduation to on-device tiers (latency_profile.json).

4.2 AGA vs the fixed ladder

AGA matches accuracy at far lower labeled-outcome cost:

dataset	oracle	AGA(LODO)	AGA_acc	fixed_acc	AGA_cost	fix_cost
ag_news	ml	model_zeroshot	0.853	0.895	0	5000
atis	ml	ml	0.848	0.848	1000	1000
banking77	ml	model_fewshot	0.807	0.870	40	5000
cifar10	model_zeroshot	ml	0.327	0.327	1000	1000
clinc150	ml	ml	0.838	0.838	5000	5000
codelangs	ml	ml	0.986	0.986	500	500
dbpedia14	ml	ml	0.968	0.968	10000	10000
emotion	ml	model_zeroshot	0.632	0.818	0	5000
esc50	ml	model_fewshot	0.050	0.283	8	100
hwu64	model_fewshot	ml	0.805	0.805	5000	5000
imdb	model_fewshot	model_fewshot	0.958	0.853	40	5000
rotten_tomatoes	model_fewshot	model_zeroshot	0.930	0.723	0	5000
sms_spam	model_fewshot	ml	0.945	0.945	2000	2000
snips	ml	ml	0.983	0.983	500	500
sst2	model_fewshot	model_zeroshot	0.968	0.770	0	5000
trec6	ml	model_zeroshot	0.795	0.855	0	5000
tweet_emotion	model_zeroshot	ml	0.613	0.613	2000	2000
tweet_hate	model_zeroshot	model_fewshot	0.640	0.560	40	250
tweet_offensive	ml	model_fewshot	0.730	0.775	40	2000
twenty_newsgroups	ml	ml	0.660	0.660	10000	10000
yahoo_answers	model_zeroshot	ml	0.670	0.670	10000	10000

LODO oracle-match: 7/21 = 0.33

Mean accuracy AGA 0.762 vs fixed-ladder(ML_PRIMARY) 0.764

Mean labeled outcomes AGA 2246 vs fixed-ladder 4017

AGA keeps LLM as terminal (skips ML training): 10/21

AGA picks rule/ML over not-better LLM (avoids perpetual cost): 6/21

4.3 Honest negative: cross-site prediction does not yet generalize

A gradient-boosted predictor of the best tier from early-observable characteristics scores **below the majority baseline** leave-one-dataset-out at 21 datasets, and its apparent skill swings with test-set noise (0.62→0.41 between n=200 and n=400 before significance-aware labeling). **Conclusion:** per-site measured routing

is what works today; cross-site prediction needs a much larger fleet. This is a finding, not a failure — it tells practitioners to measure, not guess.

5 Preliminary: image and audio

The gate is modality-agnostic (it consumes (decision, outcome) pairs); only the LLM input path is modality-specific. Rule+ML transfer to CIFAR-10 and ESC-50. The MODEL tier is preliminary: Claude vision on CIFAR images reaches 0.78 zero-shot (but our pixel-logreg ML is a weak baseline), and a spectrogram→vision proxy for ESC-50 **fails** (0.017, below chance) — native-audio models are needed. We report these as preliminary evidence of modality-generalizability, not as headline results.

6 Limitations

- **Classical-ML baselines are the cheap day-N head** (TF-IDF / pixel / MFCC + logistic regression). **Robustness check (Threat-1 ablation):** a much stronger classical tier — frozen sentence-transformer embeddings (all-MiniLM-L6-v2) + logistic regression — was run on a representative text subset. The “LLM wins on short-text sentiment/moderation” result *survives*: sst2 strong-ML 0.80 vs LLM 0.98; rotten_tomatoes 0.74 vs 0.94; tweet_hate 0.53 vs 0.77 — so it is **not** a weak-baseline artifact. High-cardinality intent still favors classical ML (banking77 0.89, clinc150 0.87, both > LLM). Caveat: frozen embeddings are not uniformly stronger (imdb 0.78 < TF-IDF 0.85 due to MiniLM truncating long reviews); a *fine-tuned* transformer ablation is the remaining future work. (Image ML remains a weak pixel-logreg; a CNN/ViT baseline is needed before drawing image conclusions.)
- Latency/cost are measured on a sample (latency_profile.json); LLM latency is network-bound and will vary.
- **Single LLM (Claude Haiku), single prompt** — accuracies are prompt/model sensitive.
- **Multimodal MODEL tier is a proxy** (spectrogram/frames); native audio/video models are future work. Spectrogram-vision is shown to fail on environmental audio.
- **Cross-site meta-prediction does not generalize** at this fleet size (§4.2).
- **n=400 test subsets** → accuracy CIs $\approx \pm 0.04$; we use significance-aware tie bands accordingly.

7 Related work

FrugalGPT / LLM cascades (cost routing); learning-to-defer (per-example deferral); champion-challenger + shadow deployment (MLOps); McNemar / Dietterich. AGA differs by choosing the *terminal* tier per stream from characteristics and graduating off the LLM under a non-inferiority gate — not per-example routing or a hand-tuned cascade.

8 Reproducibility

One command (`scripts/reproduce_dmlr.sh`) regenerates every number; MODEL stages auto-skip without an API key. See REPRODUCE.md.